Avinash Agarwal

Civilization, AI, Healthspan, Cosmos

Mathematics | Engineering | Superintelligence | Systems

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Summary

Engineering veteran with **8+ years of industry leading experience at Meta and Microsoft**, passionate about tackling civilizational challenges, especially Healthspan and Space exploration! Proven track record of driving multi-million-dollar cost optimizations, architecting highly available distributed systems, and leading cross-functional engineering teams.

Experiences

Meta

August 2025 — Present

Full-time

Menlo Park, California, USA

- **Applied AI & LLM Evaluations:** Drafted into the Applied AI organization to create model-evaluation tasks for Meta's next-gen LLM codenamed Avocado, under Meta's Superintelligence Labs.
- **AI for Healthspan:** Authored a visionary project proposal within the Applied AI org to process vast genetic datasets; Advancing milestones like Google's AlphaFold, the proposal leverages modern AI to establish novel health benchmarks and accelerate cures for aging, rare-diseases, etc.
- **Strategic Cost Optimization:** Authored a comprehensive technical proposal for WhatsApp-based authentication, proposing an architectural shift that is projected to save the company over $30 million.
- **Meta.ai Platform Reliability:** Stabilized login flows for the Meta.ai application by systematically resolving complex, intermittently failing cross-platform test suites, significantly improving CI/CD reliability.
- **Stability Lead**: Championed platform reliability operations by actively triaging and resolving critical alerts, identifying and filing tasks for key SLI/SLO coverage gaps, systemic propagation delays, etc.
- **On-Call Operations:** Completed onboarding and actively participated in team on-call rotations to rapidly triage and mitigate incidents.
- **Automated On-Call Agent:** Enhanced an alert summarizer agent by resolving prompt completeness errors; Collaborated directly with the company-wide On-Call platform owners to include this agent in their workflow, significantly improving engineering productivity.
- **Architecting Global Identity Systems:** Ensuring high-availability and security for identity systems processing billions of requests across the entire Meta ecosystem.
- **Secured the runner-up position in an internal Hackathon:** Demonstrated high engineering velocity by rapidly fixing a high volume of issues while triaging improvements for the hackathon platform.

Microsoft Azure Core

Feb 2022 — July 2025

Full-time

Redmond, Washington, USA

- **Azure Resource Graph**
  - Collaborated across engineering teams to migrate a streaming data pipeline (1B+ daily notifications) to its dedicated scale unit — Involved sophisticated design, careful cross-team planning and execution.
  - **Root-caused and championed increasing hash cardinality** across the stack, leveraging data-driven demonstrations to significantly improve distribution granularity and resolve partition skewness.
  - **High-Performance Serialization:** Championed the adoption of **System.Text.Json** across the service stack, leveraging its high-performance, low-allocation APIs to achieve significant reductions in CPU overhead and memory pressure during large-scale telemetry processing.
  - **Memory Optimization:** Championed selective caching of internal columns informed by customer query telemetry to significantly reduce active memory footprint.

- **Engineered data ingestion components** hosted on the Azure Synapse platform to support dynamic time-window loading for security datasets.
- Optimized complex SQL queries executed on Azure Synapse to efficiently load and merge multiple continuous data streams.
- Profiled and optimized high-contention services eliminating bottlenecks and achieving **nearly 10%** improvement in compute, memory and network efficiency.
- Improved on-call efficiency with multiple improvements to internal monitors, documentations, etc.

- **Azure Resource Change Analysis**
  - Delivered General Availability for the Change Analysis back-end by successfully deploying the service and all related dependencies to the US and China Government Clouds, ahead of an important Gartner cloud review.
  - Collaborated extensively to deploy the service and all related dependencies to secret Air-Gapped clouds.
  - Significantly reduced partition skewness in data-ingestion pipelines, achieving **over 30%** improvement in compute, memory and network efficiency.
  - Led the architectural evolution, including metadata enhancements and optimizations to the Change Analysis payload with extensive collaboration across teams and disciplines.
  - Delivered General Availability for the Change Actor functionality, enabling customers to audit **WHO** initiated a change and **WHICH client** (Portal, CLI, SDK) was used to make these changes.
  - Delivered General Availability for the Change Analysis front-end (live on the Azure portal) by improving Lighthouse scores, automating deployments and tests, completing A11y fixes etc.
  - Closely collaborated with the Change Safety team to reference Change Analysis resources.

Microsoft Azure Global

Feb 2022 — July 2025

Full-time (supervised by Arun Ramamurthi)

Hyderabad, Telangana, India

- **Scaling for GRAB (1B+ course completions)**
  - **Performance Profiling & Load Testing:** Performed intensive load testing and performance profiling using Apache JMeter, ensuring platform stability for massive-scale deployments across Southeast Asia.
  - Identified and contributed a fix (GitHub Issue #46) for Azure Load Testing + Terraform setup, effectively adding Azure-Load-Testing to the multi-cloud cohort supported by Terraform;
  - **System Optimization & Throughput:** Optimized core services to achieve a **20x throughput improvement** by studying and refining SQL query execution plans, eliminating redundant network calls, and implementing strategic caching layers.
  - **High-Scale Data Migration:** Architected APIs to execute the **migration of 160M+ course-completion records** from AWS to Azure ensuring data integrity during the transition.
  - **Technical Leadership & Design:** Spearheaded technical design reviews and PM alignment, advocating for extensible solutions to avoid technical debt from short-term fixes.

- **UNICEF Learning Passport (powered by Microsoft Community Training)**

  "Education is the ultimate 'Great Filter' prevention mechanism"

  - **Offline Innovation (Hackathon Lead):** Spearheaded a hackathon project to host the entire cloud service locally on Windows machines. The platform was recognized as one of **TIME's Best Inventions of 2021** for providing continuous education to children in conflict zones and refugee settings.
  - **Global Societal Impact:** This "offline-first" prototype was the key technical catalyst that prompted **UNICEF** to launch the initiative using Microsoft Community Training for no-connectivity regions, contributing to a high-scale initiative that reached **7.4 million learners across 43 countries** by 2024;
  - **Strategic Architectural Leadership:** Championed and worked on containerization and migration to **cross-platform .NET Core**; this move was critical for the platform's ability to run on evolving hardware in emerging markets.

- **Operational Support & Edge Strategy:** Facilitated the integration of **Azure IoT Hub** to support remote management and observability for offline learning hubs; assisted in testing data-sync on diverse Linux-based hardware to ensure reliable curriculum delivery in remote field operations.

- **Swachh Bharat Mission (powered by Project Sangam)**
  - Directed an engineering team through the end-to-end implementation of key mobile features, culminating in the successful launch of the official Swachh Bharat Mission eLearning mobile app.
  - Spearheaded the rapid triage of launch-critical issues in close collaboration with Aditya Agarwal and product management teams to ensure a smooth migration of the training platform.
  - Architected and implemented a high-impact search-enabled dropdown to resolve a critical UI bug; this allowed users to efficiently navigate and select from **the hundreds of cities where the platform was deployed**.
  - **Contributed to a high-scale social impact project** that successfully trained and empowered over **110,000+ municipal functionaries** across **4,000+ cities** in India.

- **BCDR & Global Traffic Management:** Designed and Implemented a mission-critical Business Continuity and Disaster Recovery architecture using Azure Front Door (already benchmarked by LinkedIn), significantly improving page-load latencies, establishing a resilient global load-balancing layer and hardening the platform's security posture for massive-scale international traffic.

- **Strategic Architectural Leadership:** Formally championed Azure Lighthouse over Azure Managed Applications to resolve cross-tenant visibility constraints; authored deep-dive technical evaluations demonstrating how Lighthouse's automation via Azure Marketplace would have accelerated time to market and eliminated multi-tenant architectural debt for **ALL of Azure Global**.

- **Identity & Authentication:** Streamlined the SMS-based login flow by implementing custom Identity Server solutions over Azure Cosmos DB and closely collaborated with Telesign to significantly reduce SMS code-sending costs across many countries; Engineered workflows for load-testing authentication flows using locally signed JWT tokens.

- **Search Infrastructure & Optimization:** Designed and implemented a custom search index featuring advanced schema optimizations and Lucene-based retrieval; Used Azure Cognitive skills to index relevant content from unstructured file formats like PDF, Office, Videos, Audios, etc. Used Azure Cognitive Search to improve search relevance with pretrained semantic analyzers like fuzzy matching, edit distance, synonyms, phonetics, etc.

- **Media Security & DRM:** Authored a technical proposal to modernize content protection for temporal media using industry-standard DRM protocols like Digital Rights Management including **Widevine, PlayReady, and FairPlay** beyond just AES-128 based encryption-in-transit.

- **Platform Localization**
  - Initially developed comprehensive support for **right-to-left (RTL) formats** before assuming full platform localization ownership.
  - Spearheaded global accessibility by **engineering an internal localization framework** and integrating with manual translation pipelines.
  - Initially developed comprehensive support for **right-to-left (RTL) formats** before assuming full platform localization ownership.
  - Conducted a **comprehensive investigation on ligature support** to ensure accurate rendering of complex scripts across the ecosystem.
  - **Custom PDF Certification Engine:** Architected a templatized HTML-to-PDF generation feature by defining rigorous localization and security requirements; strategically collaborated with IronPDF to accelerate time to market.

- **Notifications Infrastructure**: Onboarded via client-side Xamarin APIs before assuming full ownership of the platform's notification subsystem (under the mentorship of Rohan Badlani)

- **High-Throughput Bulk Ingestion:** Architected and built an end-to-end bulk upload system enabling parallel ingestion of multi-format media; significantly reduced administrative onboarding time and **super-charged the manual testing experience** for everyone.

- **Developed a scalable monitoring infrastructure** using Event Hubs and Azure Data Explorer to centralize telemetry and enable diagnostic and analytic dashboards.
- **Automated end-to-end deployments and CI/CD pipelines** enabling Public Preview release of the offering on Azure's Marketplace.
- **Advanced Thread Modeling:** Conducted deep-dive threat modeling and attack surface analysis using Microsoft internal security frameworks; ensuring the integrity of user data and platform content.
- **Privacy & Compliance:** Enabled automated compliance tooling to automate the identification of secure, licensed dependencies; ensuring rigorous adherence to global data protection and privacy standards.
- **State Persistence & User Analytics:** Developed features for real-time tracking of user progress, including "most recently viewed lesson" and "last watched timestamp" for temporal media content ensuring a seamless cross-device learning experience.

Microsoft Hackathon

2018 — 2020

Engineer

Hyderabad, Telangana, India

- Kool AI(d) for Gamers: Fine-tuned pre-trained computer vision models to extract real-time HUD metrics and game states from live Dota 2 streams; Collaborated in a team of two.
- **Koev Halev:** Actively contributed at the Microsoft IDC campus, providing domain expertise in cross-functional design discussions for a project focused on behavioral health support.

Microsoft Code.Fun.Do

2018 — 2019

Engineering Lead and Pizza Logistics Moderator

India

- **Event Moderator:** Delivered talks on project-submissions, technical-guidelines etc. and managed ad-hoc troubleshooting and Pizza logistics at **NIT Surathkal** and **BITS Pilani (Hyderabad)**.
- **Competitive Programming:** Proposed supplementing project-based events with structured programming contests (Google Kickstart, Google Code Jam, Meta Hacker Cup, ACM-ICPC, etc.), positioning rigorous algorithmic proficiency as a **foundational pillar for engineering talent.**

The Garage, Microsoft

June 2018

Engineer

Hyderabad, Telangana, India

- **IoT Home Automation:** Collaborated within a team to retrofit and control legacy electric switches with motorized 3D-printed actuators via a Raspberry-Pi based IoT layer.

Goldman Sachs

September 2016 — October 2016

Engineering Lead

Bengaluru, Karnataka, India

- **National Runner-Up (GS Quantify 2016):** Led a 3-member team in this flagship interdisciplinary event, serving as the primary contributor for the advanced algorithmic problem-solving track.
- **Data Science** (ft. Samiksha Gupta): Contributed to the data-science track for bond-liquidity-prediction.
- **Quant** (ft. Shivam Tyagi): Directed the quantitative solution track to finalize the submission.
- **Executive Presentation:** Earned a fully sponsored invitation to their Bengaluru headquarters (hosted at the prestigious Hilton Embassy) where the team delivered the concluding presentation.

Microsoft Foundry

May 2016 — July 2016

Engineering Internship

Hyderabad, Telangana, India

- **Conversational AI:** Engineered chat-bots using the **Microsoft Bot Framework**, architecting custom conversational flows for automated form-filling, within a cross-functional team.
- **Computer Vision:** Processed images using **OpenCV** to better highlight textual information, significantly improving downstream OCR engine performance.
- **Natural Language Processing:** Tested and evaluated experiences built with **Microsoft LUIS**.
- **Product Presentation:** Together with the team, delivered a highly acclaimed final product presentation to a large Microsoft Foundry audience.

Xerox Research Centre Dec 2015 — Jan 2016
Research Intern (AI/ML) Bengaluru, Karnataka, India

- **Xerox Research Innovation Challenge:** Engineered predictive neural network models using vitals data to predict patient mortality and health complications among ICU patients.
- Worked on synthetic data generation and an ensemble of models to predict various health complications based on both vitals data and lab results.
- Completed a Winter School on predictive data-science under the supervision of Vaibhav Rajan.
- Secured **third place** (online-round) and **seventh place** (onsite) in the challenge, alongside a 3-person team; successfully competing against dedicated Ph.D. research teams.

Accomplishments

- **ACM ICPC 2016:** Qualified for and represented IIT (BHU), Varanasi at the Asia-Amritapuri regionals
- **ACM ICPC 2015:** Qualified for and represented IIT (BHU), Varanasi at the Asia-Amritapuri regionals
- Ranked **15th** internationally (2nd in India) in **Codeforces Round #410 (Div. 2)**
- Ranked **83rd** internationally in **Google Kickstart Round D 2017**
- Ranked 41st in IIT Kanpur's Monthly Programming Contest in a team of 3 among 1117 other teams
- Ranked 16th on HackerEarth's October Sky among 800 international candidates
- Ranked 25th in IIT Bombay's TechFest-15 International Coding Challenge Final Round
- Ranked 12th in IIT Bombay's TechFest-15 International Coding Challenge Screening Round
- Ranked **19th nationally** in CodeChef May Challenge 2015, reaching an **overall India Rank of 31** in CodeChef's Long Challenge track.
- Reached an All India Rank of 15 in the prestigious CodeChef Long Challenge
- **IndiaHacks 2016** (ft. Sartaj Singh): Secured a **14th rank** against 3,000+ other teams in their data-science track; Invited to the on-site finals held at **Vivanta by Taj** in **Bengaluru, KA**

Projects

- **2D Physics Game Engine:** A gravitational rigid-body simulation in Java:
  - Engineered a physics engine featuring multi-object **collision detection and kinematic interactions**
  - Engineered a **Near-Earth fixed-gravity model** to simulate realistic downward acceleration.
  - Engineered a **Newtonian** $1/r^2$ **gravity model** to simulate **N-body interactions** and **galactic mergers** in empty space.
- **Multi-Agent Environment Simulation:** Built a simulation of independent RL-Agents coordinating with each other to efficiently collect items distributed in a two-dimensional grid.
- **AIMA-CSharp Open Source Contribution:** Contributed to the AIMA-CSharp repository by implementing pseudo-code in C# from Peter Norvig's *Artificial Intelligence: A Modern Approach*.
- **Image Steganography:** Implemented a system to encrypt and embed hidden messages within images.
- **Intel E5200 CPU Overclocking**: Successfully overclocked an Intel Pentium E5200 processor from a base clock of **2.5GHz** to **2.8GHz**. Involved tuning **FSB frequencies, CPU multipliers, and voltage settings**; Benchmarked thermal output and system stability under peak loads.
- **The Chess Game:** Developing an evolving chess engine from pre-college through my final year, transitioning from a foundation in Java to a sophisticated AI-driven system. Independently integrated an AI engine utilizing **Mini-Max** and **Alpha-Beta pruning**.
- **Tic-Tac-Toe AI:** Implemented a decision-tree based AI and multi-player for Tic-Tac-Toe; the computer being unbeatable in the hard difficulty level.

SKILLS

- **Leadership** — Prioritization, Cross-functional Collaboration, Reviews, Model, Coach, Care, etc.
- **Artificial Intelligence** — Reinforcement Learning (Generative Adversarial Play, Exploration vs Exploitation, etc.), Deep Learning (Neural Networks, CNNs, RNNs, Transfer Learning, etc.), Machine Learning (Mathematical Modeling and Search Heuristics), Data Science (Statistics, Classification, Regression, PCA etc.), LLM Applications, etc.
- **Engineering** — Business Continuity & Disaster Recovery, Data Structures and Algorithms, Distributed Systems and High Performance Computing (Billions of daily notifications), Data Pipelines, Authentication and Identity Systems, Blockchain, API design and Smart Contracts, IaaC, etc.
- **Security** — Ethical Hacking, Capture the Flag events, Threat Modeling, etc.
- **Privacy** — Encryption, CCPA, GDPR, etc.
- **Site Reliability** — Business Continuity and Disaster Recovery, On-call and Incident Response etc.
- **Data & Storage** — Redis, NoSQL, SQL, Blockchain, etc.
- **Search** — Apache Lucene, Elasticsearch etc.
- **Cloud & Infrastructure** — Microsoft Azure (Expert), AWS, Kubernetes, CDN, CI/CD & Infrastructure Automation, etc.
- **Frameworks & Libraries** — JVM, .NET, TensorFlow, Keras, Pandas, Scikit-learn, XGBoost, OpenCV, Unity, React, ARM, IronPDF, etc.
- **Hardware** — Semiconductors, Assembly, Firmware, etc.
- **Programming Languages** — Logo, Basic, Java, C, C++, C#, Python, TypeScript, PowerShell, Hack, etc.
- **Natural Languages** — English, Hindi, Sanskrit, Marwari, etc.

EDUCATION

Indian Institute of Technology (Banaras Hindu University) July 2013 - May 2018
Integrated Master's degree (5 years): Mathematics & Computing Varanasi, Uttar Pradesh, India

- **Thesis:** Deep Learning based Classification of Handwritten Mathematical Symbols. (Supervised by Tanmoy Som)
  - Trained and fine-tuned transfer learning models to achieve state-of-the-art accuracy on the HasyV2 dataset.
  - Successfully defended thesis through a detailed technical presentation on CNNs, transfer-learning methodologies and classification results.
  - Technologies Used: Pandas, AWS, Numpy, Tensorflow, Keras, CNN, MobileNet, DenseNet, ResNet, Wide-ResNet, NASNet, Scikit-learn, etc.
- **Teaching Assistant** for the freshman course on Mathematical Methods.
- **Problem Setter & Moderator – Hackerrank (Technex'16):**
  - **Mathletics:** www.hackerrank.com/mathletics-technex16
  - **Prayaas:** www.hackerrank.com/prayaas-technex16
- **Ranked 1st in CodeDef-2016**, a coding contest organized under Udyam, a college festival organized by the Electronics branch.
- **Ranked 2nd in Analiticity**, a campus-wide data-science contest in Technex '16.
- **Ranked 1st in COPS-OPC-2015**, a programming contest organized by the club of programmers at IIT Varanasi.
- **Microsoft Code.Fun.Do** Collaborated in a team of 4, developing a 2D top-down space shooter game using the Unity game engine.
- **Matheletics:** Ranked 3rd in the Mathematics and Computing event, in Technex-15.

- **Predicting Tourism Trends:** Participated in a team of 3 to build machine learning models to forecast tourism trends (Supervised by Santwana Mukhopadhyay)
- **Rediscovering Darkness:** Stood 2nd in an Astronomy Quiz hosted by the Astronomy Club!
- **KashiYatra-14:** Runner's up in Treasure Hunt, participated in a team of 3.
- **The Chess Game:** Stood 2nd in Technex-14 Mod-Ex; Developed an evolving chess game from pre-college, featuring a chat room (UDP, TCP, etc.)
- **Robo-Soccer:** Runners-up out of 100+ teams at the Robotics-Club's Technex-13 event; conceptualized electronic circuit designs for a D-Pad controller to operate an All-Wheel-Drivetrain.
- **Event Organizer & Moderator**: Competitive tournaments for Age of Empires, Rise of Nations, Call of Duty, Need for Speed, and Defense of the Ancients!

coursera Coursera.org

Specializations and Certifications — April 2018 — March 2025, Online

- **DeepLearning.AI:** Deep Learning Specialization
- **University of Alberta & Amii:** Reinforcement Learning Specialization
- **The State University of New York:** Blockchain Basics and Smart Contracts
- **DeepLearning.AI:** Generative AI For Software Development

FIITJEE Forum for IIT-JEE

Joint Entrance Examination — April 2011 — June 2013, Dhanbad, Jharkhand, India

- **All India Rank 494** in Joint Entrance Examination — 2013, among 1.3 million candidates.
- **All India Rank 1672** in IIT-JEE (ADVANCED) — 2013, among 150k candidates.
- **Stood 1st** in the batch of 2013, among 200+ other candidates.

Delhi Public School

Central Board of Secondary Education — June 2011 — April 2013, Dhanbad, Jharkhand, India

- **Scored 96.5%** in the All India Senior School Certificate Examination (AISSCE) — 2013!
- **All India Rank 6355** in National Eligibility cum Entrance Tests (NEET) — 2013, among 650k candidates
- **Awarded DST INSPIRE Science Scholarship for ranking in the top 1%** of AISSCE — 2013
- **State Rank 1 (Medicine)** in Jharkhand Combined Entrance Competitive Examination — 2013
- **State Rank 2 (Engineering)** in Jharkhand Combined Entrance Competitive Examination — 2013
- Recognized in the **top 0.1% of AISSCE — 2013 for excellence in Physical Education**
- Drafted into the advanced mathematics program **Aryabhatt** and represented the school at IMO!
- **Labs:** Biology, Chemistry, Physics!

De Nobili School, Sijua

Indian Certificate of Secondary Education — March 2000 — March 2011, Dhanbad, Jharkhand, India

- **Scored 87.9 %** in the Indian Certificate of Secondary Education — 2011 examination.
- **Received 90%+ in FTRE Scholarship** based on competitive national-level performance among 100,000+ candidates!
- **Practicals:** Reading, Speaking, Writing, Physical-Training, Art & Craft, Computers, Chemistry, Biology, Physics, Topology, etc.

PURSUITS

- **Metaverse** — Spatial Computing, AR/VR, WIP!
- **Science** — Medicine, Anatomy, Hardware Assembly, Theoretical Physics, WIP!
- **Multiverses** — MMORPG, RTS, FPS, Podcasts, Documentaries, Books, Anime, WIP!
- **Music** — Rock, Pop, EDM, MeaningWave, LoFi, Rap, etc.
- **Guitar** — Beginner (Eartuning, Staff Notation, Electric, Yousician, etc.)
- **Foodie** — Moderate!
- **Cooking** — Beginner!
- **Badminton** — Beginner!
- **Swimming** — Newbie!
- **Traveling** — Stars, NASA, SpaceX, Amazon Leo, Auroras, Universities, Museums, Monuments, etc.
- **Philosophy** — Ontology, Epistemology, Causality, Agency, Teleology, Teleonomy, etc.
- **Natural Languages** — Mandarin, Spanish, Punjabi, Korean, Tamil, Japanese!